

① Degeneracy:

• for each $l: 2l+1$ (m -states)

$$l = 0, n-1$$

$$\sum_{l=0}^{n-1} (2l+1) = 2 \sum_{l=0}^{n-1} l + \sum_{l=0}^{n-1} 1$$

$$2 \sum_{l=0}^{n-1} l = 2 \left(\frac{(n-1)(n)}{2} \right) = n^2 - n$$

$$\sum (2l+1) = n^2 \quad \checkmark$$

② $|C_0|^2 \int_0^\infty dr \, r^2 e^{-2r/a_0} = 1$

$$t = \frac{2r}{a_0} \Rightarrow dt = \frac{2}{a_0} dr$$

$$|C_0| \int \frac{a_0}{2} dt \left(\frac{a_0}{2} \right)^2 t^2 e^{-t} = 1$$

$$\Gamma(z) = n! = \int_0^\infty dt \, t^n e^{-t} =$$

$$|C_0|^2 \left(\frac{a_0}{2} \right)^3 (2) = 1$$

$$C_0^2 = 2^2 a^{-3} \Rightarrow \underline{C_0 = 2 a^{-3/2}}$$

